

Blast disease of Rice

Magnaporthe grisea Blast

	Prevention	Monitoring	Direct Control	Direct Control	Restrictions
 Rice field with blast symptoms on the leaves (E. Moses, Crops Research Institute)	- Use certified disease free seeds for planting or consider using tolerant varieties (eg. Amankwatia, Jasmine and AGRA rice) - Avoid excessive use of nitrogen fertilizer as this can increase the severity of the disease - Avoid drought stress to plants and extended drainage period as this makes the plants more susceptible to the disease - Practice continuous flooding to limit blast development	- Monitor plants by observing leaves for appearance of major symptom - diamond shaped lesions with grey centre and brown or reddish borders and start post planting control measures - Presence of blast lesions just before booting often dictates decisions for chemical treatment	Remove leaves with lesions or symptoms if they are not many during monitoring	When using a pesticide or botanical, always wear protective clothing and follow the instructions on the product label	
 Blast lesions on leaves (Donald Groth, Louisiana State University AgCenter, Bugwood.org)				Do not use chemicals with the same mode of action year after year as this can lead to resistance	
 Rice stem nodes showing symptoms of rice blast (Donald Groth, Louisiana State University AgCenter, Bugwood.org)				- Always consult the most recent list of registered pesticides of MOFA, Ghana - Apply Maneb (80%) (Trimangol 80WP) Apply 70g/knapsack; Broad spectrum protectant fungicide with a multi-site mode of action (FRAC group M3)	WHO Class U (Unlikely to present acute hazard in normal use) Start application when lesions first appear and just before booting to prevent neck blast. Repeat application at 10days interval.Observe PHI of 14 days and : DO NOT enter into treated areas until the spray has dried unless wearing cotton overall.
				Apply Kingstar WG (Azoxystrobin (60%)+Cyproconazole (24%)) Apply at a rate of 75ml/15l knapsack; FRAC groups: azoxystrobin C3 & cyproconazole G1	WHO Class II (Moderately hazardous), Apply at first sign of disease attack, usually from tillering to early flowering. Ensure thorough coverage of application. Repeat application upon monitoring at early booting stage. PHI is 15 days. DO NOT enter into treated areas until the spray has dried unless wearing cotton overalls

Ghana

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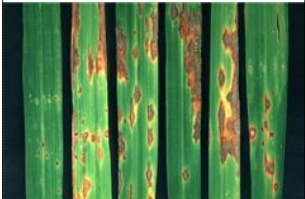


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Management of Rice blast

Pyricularia oryza, *Magnaporthe grisea*

	Prevention	Monitoring	Direct Control	Direct Control	Restrictions
 Rice blast. Photo by IRRI (CC BY-NC-SA)	• Deep ploughing with organic matter • Use of disease/spot free seed • Avoid the use of excessive N and apply balanced fertilizer • Weed control to remove alternate host plants	• Regular visits and observation of the rice plants immediately if low night and high day temperature, drizzly rain, morning dew and windy weather • Look for angular shaped lesions (eye shaped) with pointed ends on the leaves • Look for infected nodes, panicle and neck • Observe alternative host grasses for symptoms • Late wet and dry season is favorable • Sandy loam soil and dry condition also favour the disease	• Spraying of MOP (@6g/L) and elemental sulphur (@6g/L) during PI-boot stage • Split application of nitrogen fertilizer do not exceed recommended dose • Keep soil moist and apply irrigation during drought	• Apply fungicides during early morning and or late afternoon • Use mask during spraying of pesticides • Don't apply fungicides high concentrations of chemical or more than two times	
 Lesions on leaves are elliptical or spindle-shaped. Photo by Chin Khoun Min				• Application of fungicide during boot and after flowering is recommended Spray premixed Tebuconazole +Trifloxystrobin usually @ 0.5g/L of water • Spraying of powder formulation of Tricyclazole @ 0.75g/L of water • Spraying premixed Tricyclazole and Propiconazole- @ 2-3 ml/L water	• Trifloxystrobin- WHO U , FRAC 11; Tebuconazole- WHO II , FRAC 3 (Use two sprays 7-10 day intervals apart) • WHO II FRAC 16.1 (Use two sprays 7-10 day intervals apart) • Tricyclazole- WHOII, FRAC 16.1; Propiconazole- WHO II, FRAC 3 (Use two sprays 7-10 day intervals apart)

Bangladesh

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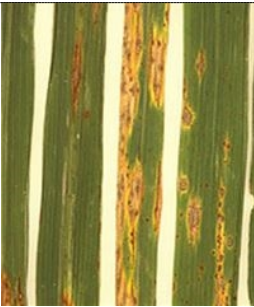


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Rice blast - Kenya

Magnaporthe grisea (Pyricularia grisea) Mpunga/Mchele-Swahili

 Symptoms of rice blast (Photo: R.K. Webster, ARS-USDA)	Prevention - Plant resistant varieties such as Nerica 1, 4, 10 and 11 for upland areas; SC 213, NIBAM 108, NIBAM 109, NIBAM 110, IR-025, and N221. - Plant early to avoid late-season blast pressure. - Flood the field as often as possible. - Avoid excessive use of nitrogenous fertilizers. - Destroy infested crop residue before planting. - Practise proper water management.	Monitoring - Examine plants in several locations throughout the field for the presence of leaf lesions. - Scout for diamond-shaped elongated lesions with a gray or white center and brown or reddish brown border on the leaves, nodes, panicles and collar of the flag leaves. - Look out for a bleaching whitish colour on the head on the lower portion of the panicle. - Look out for a rotten panicle neck caused by a broken neck condition.	Direct Control Burn or compost diseased tissues.	Direct Control - When using a pesticide, always wear protective clothing and follow the instructions on the product label - Do not use chemicals with the same mode of action year after year as this can lead to resistance - Always consult the most recent list of registered pesticides (PCPB) <i>Trifloxystrobin</i> 100 g/L + and Tebuconazole 200g/L. NATIVO SC300. Rate: 5-10ml/L - Difenoconazole (Score 250 EC). Rate: 20ml in 20L	Restrictions <i>Trifloxystrobin</i> WHO class U (unlikely to present acute hazard), Tebuconazole WHO II (moderately hazardous). NATIVO SC 300: Blue colour band. MRL 2.0mg/Kg trifloxystrobin 0.2mg/Kg, PHI =40 days. Tebuconazole, 2.0 g/Kg. MRL = 2.0mg/Kg - PHI 14 days. WHO class II (moderately hazardous)
 Rice leaves infected by blast. (Photo: IRRI)					



Kenya

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Coffee leaf rust

Recognize the problem

Coffee leaf rust, is a very destructive disease. It is recognised by yellow powdery spots on the underside of the leaves. Infected leaves fall off resulting in death of the branches. It reduces yield up to 70% in severe cases.

Background

Coffee leaf rust is caused by a fungus, which infects coffee leaves of all stages and makes them fall down. It is severe in dry and warm periods. Coffee leaf rust spreads mainly by wind, but human beings can also spread it when they walk in the farm and the fungus sticks to their clothes and body just like powder. Chemical control measures are costly and need to be taken when the disease has reached the threshold which is generally two spots per leaf.

Management

1. General

- Coffee should be planted at recommended spacing. It should also be pruned to open up the coffee bush in order to create unfavourable conditions for the development of the fungus and allow sprays to reach leaves which are inside the canopy.

2. Use of chemicals

- Use copper oxychloride for prevention of coffee leaf rust. Mix 50 grams of copper oxychloride in 15 litres of water, and sprayed on 20 trees after every two weeks.
- When coffee leaf rust is established, spray cyproconazole by mixing 10ml of cyproconazole in 10l of water, and spray once a year.

Note: sprays must be directed on the underside of coffee leaves.

3. Plant resistant varieties

- Plant resistant varieties when available.

Coffee leaf spots on leaves Eastern Province of Rwanda. (Photo by Stanislas Ushimiyimana)



Coffee branches which have lost almost all leaves due to coffee leaf rust. (Photo by Noah A. Phiri, CABI)



When using a pesticide, always wear protective clothing and follow the instructions on the product label, such as dosage, timing of application, and pre-harvest interval.

Scientific name(s) > *Hemileia vastatrix*

The recommendations in this factsheet are relevant to: Ghana, Kenya, Rwanda



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Coffee leaf rust

Hemileia vastatrix



Symptoms of *H. vastatrix* rust on leaf (CABI)



Rust symptoms on underside of leaves (CABI)

Prevention

- *Hemileia vastatrix* is a fungus which infects coffee trees, causing coffee leaf rust. The fungus is spread by spores between leaves, resulting in defoliation and dieback.
- Use recommended tree spacing of 3x3m to reduce humidity and fungal spread.
- Use of resistant varieties e.g. Catimor, S.73 or Bourbon.
- Use of organic manure to increase nutrients for plants, as low-nutrient trees are more susceptible to rust.
- Weeding and pruning of all unnecessary vegetation (important to do after harvest) to reduce the humid environment which supports fungal spread.

Monitoring

- Regularly monitor trees for rust fungus symptoms:
- Yellow/orange powdery spots on the underside of leaves with corresponding yellow patches on upper side. 2-3 mm diameter, will increase in size up to 4 cm.
- Centre of spots become necrotic and turn brown.
- Necrotic leaf area may spread until entire leaf is dead.
- Infection of leaves results in defoliation, this may be premature in some trees.
- Infection of tree may cause premature ripening of fruit, resulting in low quality coffee beans.

Direct Control

- Prune and burn heavily infected branches, this will open up the crop area and reduce favourable environment for fungus.

Note: Pesticides may be available to control this pest. Please check with the Ministry of Agriculture in your country to find out which pesticides are registered in your country and the local restrictions for their use.



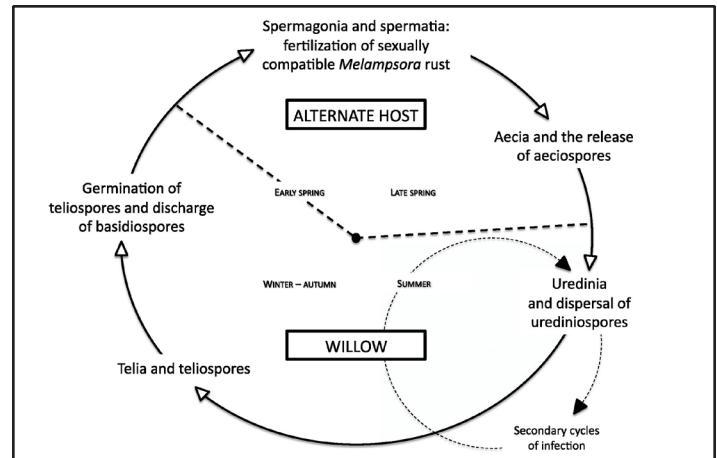
Leaf Rust

Introduction

Leaf rust of willow (*Salix* spp.), a widespread and common disease of native and cultivated *Salix*, can severely defoliate shrubs and/or trees with up to 100 percent of leaves shed six or more weeks prior to normal autumn leaf drop. The disease is caused by fungi belonging to the genus *Melampsora* (Uredinales: Melampsoraceae) and is considered the most serious disease affecting intensively managed willow plantations in North America and Europe.



Willow rust (*Melampsora* spp.): Uredinia and exposed urediniospores on the lower-leaf surface of *Salix ericocephala* (Tully, NY; August, 2009)



Willow rust (*Melampsora* spp.) life cycle

Biology

Willow-infecting *Melampsora* species have complex life histories during which they alternate between willow and an unrelated host to complete their life cycles. Two species, *M. epitea* and *M. paradoxa*, exist in N. America. Important alternate hosts of *M. epitea* in New York State are balsam fir, white fir, and saxifrages. *Melampsora paradoxa* infects willow as well as European larch, western larch and tamarack. Yellow to orange pustules (uredinia) appear on the underside of willow leaves beginning in late spring and continue throughout the summer. These pustules eventually rupture the epidermis to release large numbers of golden-yellow spores (urediniospores). This stage of the rust life cycle, referred to as the “repeating phase”, serves to continuously reinfect the initial *Salix* host as well as to spread the rust to surrounding willows. In mid-autumn, uredinia change to orange-brown or dark brown telia that overwinter on fallen willow leaves and release fragile basidiospores the following spring. Basidiospores are wind-disseminated and infect the foliage of the alternate host (e.g., balsam fir). Spermatia appear shortly after infection in the late spring and are followed by aecia containing yellow to orange aeciospores, which are dispersed by wind and infect the current growth of willow. Within two-weeks,



Willow rust (*Melampsora* spp.): uredinia bearing urediospores on the lower-leaf surface (28x) of *Salix ericocephala* (Constableville, NY; August 2009).

uredinia and urediniospores are produced on the lower surface of willow leaves; thereby, renewing the fungus life cycle. There is good evidence to suggest that special forms of *Melampsora* spp. can overwinter as mycelium or uredinia within dormant willow buds and stems. If so, this eliminates the need of an alternate host and shortens the annual disease cycle.

Susceptibility

Severe infestations of willows with *Melampsora* spp. can result in defoliation, premature leaf senescence, and predispose the host trees to abiotic stressors (e.g., competition and drought) as well as secondary disease organisms (e.g., foliar and/or canker pathogens). These compounding factors can lead to significant reductions in biomass. The susceptibility of native *Salix* species (e.g., *S. discolor*, *S. ericocephala*, *S. nigra*, *S. pyrifolia*) to *Melampsora*-infection has not been elucidated and warrants future study, however, all willows endemic to cool temperate regions of N. America are thought to be susceptible to at least one *Melampsora* species. In biomass-producing plantations in NY, the majority of *S. ericocephala* appear to be highly susceptible to *Melampsora* rust. Several *S. purpurea* cultivars as well as crosses between *S. viminalis* and *S. miyabeana* also

are susceptible to infection; yet, leaf infections on these clones are limited to the lower third of the canopy and rust-associated defoliation is rare. Further field and inoculation studies are required to quantify rust resistance among willow clones and genotypes destined for commercialization.

Management

Severe reductions in yield can be avoided by planting mixtures of willow clones that differ in resistance to *Melampsora* rust. Six to ten genotypes are commonly planted in European coppice plantations where disease pressure is high. Fungicides (e.g., triadimefon or mancozeb) also are effective at protecting willow against rust. However, regular fungicide applications often are impractical economically and environmentally. Sufficient chemical coverage for the control of rust is further complicated 1-3 years after coppicing by stool/stem-density and the intermingling of foliage. Therefore, fungicide application is not a favored method for controlling rust in short-rotation coppice plantations.

Selected References

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- Shawn C. Kenaley¹, George W. Hudler¹, Dawn Dailey O'Brien¹, Kimberly D. Cameron², and Lawrence B. Smart².

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Mango Powdery Mildew

Recognize the problem

Mango powdery mildew is a disease that attacks flowers and leaves causing a white powdery appearance (see top and bottom pictures).

Powdery mildew white dusty spores on mango flowers.



Background

- The disease is caused by a fungus.
- The fungus grows on the plant and produces spores that appear as white powder on the surface.
- It is spread through the air and is more serious in dry but humid weather conditions.
- It causes leaves and flowers to drop hence no or very little fruit formation. Young leaves are usually more diseased than old ones.
- The disease can be controlled using cultural methods and chemicals.

Management

- Cut the mango tree branches to keep them short for easy management and air flow.
- Remove diseased leaves and flowers and bury or burn them to reduce the source of the disease.
- Spray the trees with plant penetrating chemicals such as those ending in "azole" or "strobil", changing them regularly with surface chemicals such as those containing sulphur.
- Spray on weekly intervals, following the instructions on the label.
- Start spraying just before flowering and continue throughout the flowering period.

Powdery mildew white spores on mango leaves.



When using a pesticide, always wear protective clothing and follow the instructions on the product label, such as dosage, timing of application, and pre-harvest interval.

Scientific name(s) > *Oidium mangiferae*

The recommendations in this factsheet are relevant to: Kenya, Uganda



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Powdery mildew on Mango

Oidium mangiferae



Powdery mildew on inflorescence (Scot Nelson)



Powdery mildew on leaves (Scot Nelson)



Powdery mildew (*Oidium* sp.) (Florida Department of Agriculture and Consumer Services, Bugwood.org)

Prevention

- Plant-resistant or tolerant varieties.
- Plant at wider spacing from 8 x 8 m to 12 x 12m. If you plant at smaller distances, keep thinning until the standard spacing is reached.
- Practice good orchard sanitation:
- Prune tree branches to allow air circulation.
- Pickup and destroy fallen leaves and flowers.
- Remove tall weeds from orchards.
- Fertilize the orchard properly (increased rates of N-P-K application decreases the incidence of powdery mildew). Avoid late-season applications of nitrogen fertilizer.
- As the fungi is carried by the wind, intercropping with other fruit trees or forestry species reduces the spread of the disease.
- In nurseries, avoid overhead watering to help reduce the relative humidity.
- Keep all equipment clean and dry to prevent spread of the fungus.

Monitoring

- Before blossoming, observe 10 marked trees/ha in the orchard for symptoms each week:
- Young tissues (flowers, leaves and young fruits) show small patches of white powdery mycelium which can later merge to cover large areas.
- Young infected leaves may become distorted, develop grey necrotic lesions on the upper side of the leaves and tend to curl downwards.
- If severe, the leaves may become brown, dry and fall.
- Inflorescences: fail to open and may drop from panicles, can become completely covered by the mildew and blacken or become brown and dry.
- Fruits: severe blossom infection can result in complete loss of fruit. The epidermis of the infected newly set fruit cracks and forms corky tissue. Entire fruit may become yellow and misshapen, covered in mildew and fall prematurely.
- Carefully monitor the flowers as they are susceptible to infection once they have begun to open on the panicles, 3-5 weeks after bud break.
- Monitor more frequently during favourable conditions, when temperatures are mild (23°C) and air is moist (as little as 20%).
- If 2-3 trees are infested (5-10 % of leaves or flower bunches) out of 10 trees, then consider action.

Direct Control

- Remove severely infected panicles and all infected plant parts, and burn or bury to at least 60 cm deep. Do not compost infected plant debris.
- Spray infected shoots with a solution of: 6 spoons of baking powder, 3 spoons of white oil and white bar soap foam and mix with 15 litres of water.

Note: Pesticides may be available to control this pest. Please check with the Ministry of Agriculture in your country to find out which pesticides are registered in your country and the local restrictions for their use.

Tomato Powdery Mildew

Recognize the problem

This disease is commonly known as ash colour disease. It can be severe in plants grown in the green house and plastic tunnel due to poor air circulation.

Background

At first yellow spots appear on leaves, then white powder or ash. In severe cases, this white ash can be seen on all the plant parts. The spots become reddish brown at a severe stage and leaves look like they have ash on them. It is easily spread by wind. In severe disease the leaves fall off. Potato and brinjal can also be attacked.

Powdery mildew, leaves look ash coloured.



Management

- Plant crop with wide spacing to increase the air movement so that the moisture is reduced.
- Remove infected leaves when the first spots appear. Do not touch healthy plants after touching infected plants.
- Open the greenhouse and remove the covering plastic of plastic houses at the time of first disease onset.
- Spray Karathin (dinocap) @ 1ml/litre of water, 2-3 sprays at 10 day intervals, starting at first sign of disease.

Leaves die with severe infection.



When using a pesticide, always wear protective clothing and follow the instructions on the product label, such as dosage, timing of application, and pre-harvest interval.

Scientific name(s) > *Oidium neolycopersici*

The recommendations in this factsheet are relevant to: Nepal



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Stem rust of wheat

Recognize the problem

Stem rust is caused by a fungus. The fungus produces symptoms on almost all above-ground parts of the plant, especially the stem and leaf sheathes. At the initial stages of the disease, long, narrow streaks of red, dusty spots (pustules) are formed on all green parts of the plant. These spots are brick red in the beginning and become black at maturity. Moisture and warm weather favour the development of this disease, which usually appears at the heading stage. The fungus weakens the plant by taking its nutrients.

Background

In nature, stem rust disease requires a living host for its growth and reproduction and cannot exist independently. The disease arrives at a new plant after being spread by the wind, usually travelling from the hills to the plains. Stem rust is favoured by hot days, mild nights and rain or dew that causes leaves to become wet. The disease can spread to healthy plants via running water, for example rain splash or water channels. The overuse of nitrogen also favours the development of the disease.

Management

- Practise crop hygiene by removing volunteer wheat, as it can provide a green bridge for rust carryover
- Cultivate early sowing and early maturing varieties
- Avoid heavy irrigation
- Destroy diseased tillers and cereal grass weeds, especially volunteer wheat
- Avoid heavy doses of nitrogenous fertilizers
- Cautious use of potassic fertilizer helps to minimize the susceptibility of plant
- When deciding if a fungicide spray is needed, consider crop stage and potential yield loss

Stem rust of wheat. (Photo by Petr Kosina/CIMMYT (CC BY-NC-SA))



Stem rust. (Photo by Yue Jin)



When using a pesticide, always wear protective clothing and follow the instructions on the product label, such as dosage, timing of application, and pre-harvest interval.

Scientific name(s) > *Puccinia graminis*

The recommendations in this factsheet are relevant to: Pakistan

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Stem Rust on Wheat

Puccinia graminis f. sp. tritici

	Prevention	Monitoring	Direct Control	Direct Control	Restrictions
 Stem rust (B. Steffenson)	- Sow resistant varieties because they provide the best protection against stem rust. - Balanced use of fertilizer during the crop season to ensure good health of plant. - Use disease free seeds. - Ensure field sanitation from barberry alternate host plants. - Spacing and field orientation away from winds coming from infected areas may protect from wind borne sources of inoculum. - Practice no tillage or minimum tillage to reduce over wintering spore inoculum.	- Inspect wheat crops every 7 to 10 days from flag leaf emergence to early dough grain development. - Carefully inspect different plant parts, especially the lower stems, for symptoms. Spend at least 15 minutes walking through each wheat crop. - If stem rust is detected within a region, increase inspection to every 4 to 7 days. Look for reddish-brown, powdery, oblong pustules on the stems, leaves, glumes, awns and kernels. - If stem rust is detected, then collect 10 stems from 10 random locations (total 100) to determine the percentage of stem rust infection. - Whenever the disease appears it is above ETL, so inspect the field regularly to protect crop.	- Remove the green bridge, including volunteer cereals e.g. barley, common barberry (and some additional Berberis, Mahoberberis, and Mahonia spp.). - Also ask your neighbours to monitor their crop to check the disease. - Use earlier maturing wheat varieties in cooler areas downwind from warmer wheat growing areas.	- Check that the product is registered, and use the maximum recommended label rate for stem rust control in wheat. Most pesticides have protective, curative, and systemic activity. - It is important to consider withholding periods when choosing an appropriate fungicide. - It is always important to read the chemical label before use and ensure protective clothing while spraying. - Avoid repeated use of fungicides with the same active ingredient in the same season. - Tebuconazole 430 g/ L (eg. Folicur®430 SC), propiconazole 230 g/L (eg. Tilt250 EC, Bumper250 EC). - Triadimefon (e.g. Bayleton 25 WP) @ 400 gm/100L water per acre.	- Both Tebuconazole & Propiconazole: WHO Class II (Moderately hazardous). - WHO Class II (Moderately hazardous).



Pakistan

CREATED/UPDATED: June 2013/September 2022

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Wheat stem rust

Recognize the problem

Wheat stem or black rust appears in form of raised spots on leaves, leaf sheaths, heads and stems. The raised spots are usually brownish-red to red, dark brown, and black, hence the designation "black" rust. Infected areas are rough to the touch. The disease may cause over 70% yield loss in farmer's field.

Background

Wheat stem rust is caused by a fungus that can spread by plant-to-plant contact and by wind. Stem rust is favoured by hot days and adequate moisture for night time dews. Wind can effectively disperse the pathogen over small distances to cause local disease outbreaks within the crop or over great distances to cause new infections. Rain is necessary for effective establishment of the disease on the plant. The fungus survives throughout the year on infected wheat crops and volunteer wheat plants, barley and grass relatives. However, there are some varieties which can withstand the disease. The disease is also spread by people when they walk in the garden because the disease gets carried on their clothes.

Management

- Grow the following resistant varieties which are available at Rwanda Agriculture Board (RAB): KS Mwamba, E161, Bisagi, Musama and E48
- Use balanced amounts of fertilizers as follows: apply 250 kg/ha of N17P17K17 at sowing time and 100 kg/ha urea 4-6 weeks after planting.
- Remove volunteer wheat from your garden to control wheat stem rust and other diseases
- Remove plant residues, such as straw, stubble and weeds
- Do not allow many people in your farm to avoid the spread of the fungus through hands, clothing, or shoes.
- Weed regularly.
- Grow wheat in rotation with potato and beans to reduce build-up of the disease in the soil
- Prevent stem rust by spraying fungicides containing tebuconazole, and sprayed at booting stage. Mix 3 ml of the fungicide in 1 litre of water, and spray this mixture just before the disease occurs. The fungicide is available at local markets.

Stem rust symptoms on stems and leaf sheaths. (Photo by ARS, USDA)



Stem rust symptoms on leaves and sheath. (Photo by ARS, USDA)



When using a pesticide, always wear protective clothing and follow the instructions on the product label, such as dosage, timing of application, and pre-harvest interval.

Scientific name(s) > *Puccinia graminis f.sp. tritici* Eriks. E. Henn.

The recommendations in this factsheet are relevant to: Ghana, Rwanda



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Stripe Rust

Recognize the problem

Stripe rust disease is a serious fungal disease of cereals, especially wheat, which causes yield loss. Even if the number of kernels from an infected plant is high, yield will still be lower as the kernels will be smaller than those from a healthy plant.

This disease has 5 life stages. The main symptom of this disease is the development of slim lines beside each other on the blades of the leaves. The length of the lines is not always the same but the colour of the lines is always yellow to red, which is why the disease is called stripe rust.

Background

The main host for this disease is the Barberry bush. The disease is carried from one plant to another by wind, rain and humans.

This disease is favoured by temperatures of 10-20°C. If the conditions get hotter than this then the growth of the disease will stop. The disease will attack when the wheat is flowering and when the wheat is mature, the yellow lines turn black.

Management

- Remove barberry plants from the field
- Remove and destroy unwanted weeds
- Avoid from more water or less irrigation
- Cultivate the wheat in lines
- Cultivate improved seeds like wheat Gul, Solh, Pamir, Lalmi-2, Lalmi-3, Darulaman-07, Moqawom09, Chont10, etc.
- Less wheat seed cultivation
- Crop rotation with other plants like beans and vegetables.
- Chemical usage should be used as a last option. Apply sulphur powder in the first stage of the disease.

Stripe rust. (Photo by CIMMYT)



Stripe rust. (Photo by CIMMYT)



When using a pesticide, always wear protective clothing and follow the instructions on the product label, such as dosage, timing of application, and pre-harvest interval.

Scientific name(s) > *Puccinia striiformis*

The recommendations in this factsheet are relevant to: Afghanistan



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Stripe rust in wheat - Pakistan

Puccinia striiformis Yellow Rust



Close-up view of a spring wheat leaf showing signs and symptoms of stripe rust. (Source: Howard F. Schwartz, Colorado State University, Bugwood.org, CC BY 3.0)



Stripe rust inside a glume. (Source: Mary Burrows, Montana State University, Bugwood.org, CC BY 3.0)

Prevention	Monitoring	Direct Control	Direct Control	Restrictions
• Cultural Practices: Avoid early planting of winter wheat because it will get affected by stripe rust. Reduce volunteer grasses because they act as hosts to carry stripe rust season to season. Use cultivar mixtures to reduce fungal attack. Avoid excessive irrigation to reduce humidity in environment. • Mechanical practices: Destroy previous residues if infected. Remove Berberis plants because it's an alternative host for rust. • Use partially resistant varieties like Akbar-19 or Ujala-16.	• What to look for?: Tiny yellow-orange fungus spots/pustules. Small, circular spots that later develop into yellow strips. • Where to look?: Leaf, leaf sheaths, awns and inside glumes. Mostly upper leaves. • When to look and how often?: From seedling to maturation. Through-out the season, mostly in rainy season. Temp. around 15-20°C and humid atmosphere. Thrice per week when disease attack is expected. • When to consider taking action?: 10-20 leaves with symptoms at few spots in the field, or when more than 5% of plants in a field have more than 5% of their flag leaf surface infected. • Too late to take action if 40-50 leaves per many spots in a field have symptoms, most of your plants are likely infected, and action is too late.	• Preventively spray <i>Sophora flavescens</i> extracts at 10mg/ml, 30L/660 square metres, once a week from jointing to grouting stage. • Preventively spray Neem oil products at 1 fluid ounce per gallon of water. • Note that these botanicals cannot cure a plant, they can only reduce the spread.	• When using a pesticide or botanical, always wear protective clothing and follow the instructions on the product label, such as dosage, timing of application, pre-harvest interval (PHI), max. number of sprays, and restricted re-entry interval (REI). Do not empty into drains and water sources. • Apply propiconazole (e.g. Tilt™ 25% EC, Azostar™ 18.7% SC) at 200ml/acre, 120L/acre of water. Systemic action. Foliar spraying. • Apply flutriafol + azoxystrobin (e.g. TOPGUARD EQ™ FMC at 100ml/acre, 120L/acre of water). Systemic action. Ground 10 gallon/acre. • Apply bixafen + flutriafol (e.g. Lucento™ FMC at 150ml/acre, 120L/acre of water). Systemic, contact action. Apply as ground spraying.	• WHO class II (moderately hazardous). Harmful to parasitoids. Best results when disease is in early stages. PHI: 45 days. REI: 1 day. • WHO Class II. Moderately harmful to bees and pollinators. PHI: 30 days. REI: 12 hours. • Bixafen is not classified by WHO, but is highly toxic to fish, and moderately toxic to birds, aquatic invertebrates and algae. Flutriafol: WHO Class II. PHI: 30 days. REI: 12 hours. Use maximum twice per year.



Pakistan

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Stripe rust of wheat

Puccinia striiformis f. sp. *tritici* Piliya



Crop infested with stripe rust (Darrell Wesenberg, Bugwood.org)



Symptoms of stripe rust on a leaf (Howard F. Schwartz, Colorado State University, Bugwood.org)

Prevention	Monitoring	Direct Control	Direct Control	Restrictions
• Sow resistant varieties for irrigated/ unirrigated areas. • Timely sowing (first week of November). • Seed treatment with tebuconazole (2 g/kg of seed). • Apply balanced dose of NPK. • Promote cultivation of resistant varieties for fodder purpose. • Seed replacement every 3-4 years. • Sowing of mosaic of 2-3 varieties at village level. • Avoid promotion of cultivation of a single variety on larger areas. • Strictly follow irrigation schedule. • Cleaning of clothes and shoes after visiting infected field.	• Look for yellow stripes/turmeric powder (spores) on flag leaves (upper most); if there is no powder it may be spot blotch or a nutritional deficiency. • Undertake strict monitoring in water logged areas. • Monitoring after 60 days (last week of December) in irrigated areas and after 30 days in late sown areas of wheat. • Be more vigilant when dew formation takes place. • Share monitoring information at community level, village level, district level and state level. • Check crops three times a week from January onwards.		• Foliar application of propiconazole or tebuconazole at a rate of 1 ml/L of water on the appearance of symptom(s).	• WHO class II (moderately hazardous).
			• Crop raised for seed purpose, apply foliar application of Propiconazole or Tebuconazole (1 ml/L of water during first week of February).	• WHO class II (moderately hazardous). Apply at tillering stage and repeat spray after 15-day interval. However avoid spraying during rains.
			• Tilt/Folicur (Trizoles), Systemic/ FRAC.	• May harm aquatic organisms.

India

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